Energy Absorption in Cellular Materials

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Abstract

Cellular metals combine a low density with a long compressive plateau at nearly constant stress, which makes them efficient absorbers of kinetic energy. This paper brings together, in a single notation based on the rigid, perfectly plastic, locking (R–P–P–L) idealisation, the analytical results needed to design cellular-metal protection against low-velocity impact, high-velocity impact and air blast. For low-velocity impact, a selection and sizing procedure is set out that couples the damage tolerance of the protected object with charts of energy absorbed per unit volume, mass or cost, and is illustrated for an air-dropped electronic unit. For a rigid mass striking a foam column, closed-form solutions are obtained both by conserving momentum across the compaction front and by a global energy balance. The two equations of motion differ only by a factor of one half in the inertial term, and the critical column lengths that they predict are in the ratio $[1 + (1 + \phi)^{1/2}]/2$, where ϕ is the ratio of the dynamic to the plastic stress. The energy balance is therefore conservative, and the discrepancy becomes significant once the impact velocity exceeds $v^* = (\sigma_{\mathrm{pl}}\varepsilon_{\mathrm{D}}/\rho_0)^{1/2}$, approximately $73\,\mathrm{m\,s^{-1}}$ for a typical aluminium foam. Applied to sacrificial blast cladding, the analysis shows that the face-plate velocities produced by side-on air blast lie well below v^*, so that a quasi-static treatment is adequate. Finally, foams are compared with thin-walled tubes: at equal relative density a tube absorbs roughly three times as much energy per unit volume, although a foam offers isotropy and a steadier load.

Keywords: cellular metals; metal foam; honeycomb; energy absorption; impact; shock compaction; blast protection

1. Introduction

Structures that must survive large transient loads without a corresponding increase in mass are naturally built from cellular solids. In a typical metallic foam or honeycomb the solid occupies 20 % of the volume or less, which gives a high specific stiffness and strength together with a large capacity for irreversible deformation (Gibson and Ashby, 1997; Ashby et al., 2000; Wadley, 2006). The same architecture is found in wood, cork and trabecular bone, and has recently been taken to an extreme in hollow-tube nickel microlattices of relative density near 10^{-4} (Schaedler et al., 2011).

The property of most importance for protection is the response in compression. Once the cell walls begin to collapse, the stress remains close to the plateau stress σ_{pl} over strains of 0.6–0.8, until the cells are fully compacted at the densification strain ε_{D} (Gibson and Ashby, 1997; Ashby et al., 2000). A cellular metal therefore absorbs a large energy per unit volume, $\sigma_{\mathrm{pl}}\varepsilon_{\mathrm{D}}$, while limiting the stress transmitted to whatever lies behind it. This combination has led to its use in packaging, crash structures and landing systems and, increasingly, in sacrificial cladding against blast and ballistic threats (Banhart, 2001; Hanssen et al., 2002). Because inadequate protection can lead to injury or to the loss of equipment as well as to structural failure, design tools that are both reliable and easily checked are of practical value.

The analytical results needed for such design are scattered across several communities: material-selection methods based on property charts (Ashby et al., 2000), studies of shock propagation

Table 1. Methods used to join sheets in the manufacture of honeycomb cores (Bitzer, 1997).

Method	Principle
Adhesive bonding	Sheets joined by a structural adhesive; used for about 95 % of cores
Resistance welding	Current passed through the contacting sheets heats and fuses them locally
Brazing	A filler metal of lower melting point is melted into the joint
Diffusion bonding	Sheets held together at high temperature until atoms diffuse across the interface; no mass is added
Thermal fusion	The sheets themselves are locally melted and fused

in cellular media (Reid and Peng, 1997; Harrigan et al., 2010) and empirical correlations for air-blast loading (Kinney and Graham, 1985; Smith and Hetherington, 1994; Karlos and Solomos, 2013). The present paper brings these results together in a single notation built on the R–P–P–L idealisation. The manufacture of honeycomb cores is described first, since the route determines the cell geometry and hence σ_{pl} and ε_{D} (§ 2), and the mechanics of energy absorption are summarised (§ 3). A selection and sizing procedure for protection against low-velocity impact is set out in § 4. In § 5 the problem of a rigid mass striking a foam column is solved by two methods, conservation of momentum across the compaction front and a global energy balance, and it is shown in closed form how and when their predictions diverge. The energy method is extended to blast cladding in § 6, and foams are compared with thin-walled tubes in § 7. Finite-element illustrations of the deformation modes are given in § 8, and the limitations of the approach are discussed in § 9.

2. Manufacture of metallic honeycomb

The hexagonal honeycomb is among the simplest cellular topologies to manufacture, and the most widely used. It forms the core of lightweight sandwich panels and is also employed for energy absorption, flow straightening and acoustic damping (Bitzer, 1997). Any material that can be supplied as thin sheet can be made into honeycomb, from aluminium alloys, the most common metallic choice, to aramid paper and carbon-fibre composites. Adjacent sheets are joined by one of the techniques listed in table 1. Some 95 % of cores are adhesively bonded; the remaining methods are used mainly where the core will see service temperatures above about 400 °C (Bitzer, 1997).

Two routes are in common use. In the *expansion process* (figure 1), aluminium foil is unrolled and cut into sheets, onto which strips of adhesive, typically nylon–epoxy or nitrile–phenolic (Bitzer, 1997), are printed in positions that alternate from one sheet to the next. The sheets are stacked into a block, known as a HOBE (honeycomb before expansion) block, in which the adhesive is cured under pressure at elevated temperature, and the block is sliced to the required core thickness. Each slice is then pulled apart: the unbonded segments of foil yield plastically at the nodes, which fixes the hexagonal geometry permanently.

In the *corrugation process* (figure 2), the foil is passed between toothed rolls, or through a press, that form it plastically into half-hexagons. The corrugated sheets are cut, stacked node to node and joined, usually by resistance welding or brazing; curing an adhesive under pressure is unsuitable here, because the pressure would distort the pre-formed cells. Since thicker sheet can be formed in this way, and the double-walled nodes require a larger load to buckle, the corrugation route produces denser and stronger cores than expansion (Bitzer, 1997).

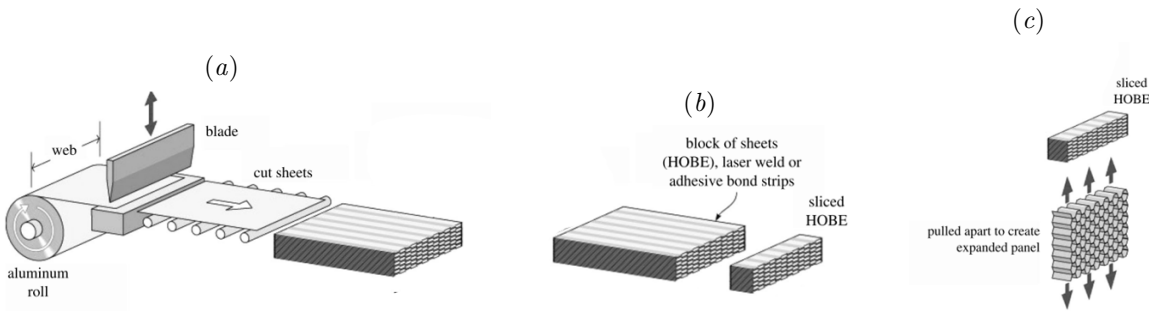


Figure 1. The expansion process: (a) unrolling and cutting of foil; (b) stacking of sheets carrying alternating adhesive strips into a HOBE block; (c) expansion of a slice into hexagonal cells. Adapted from Wadley (2006).

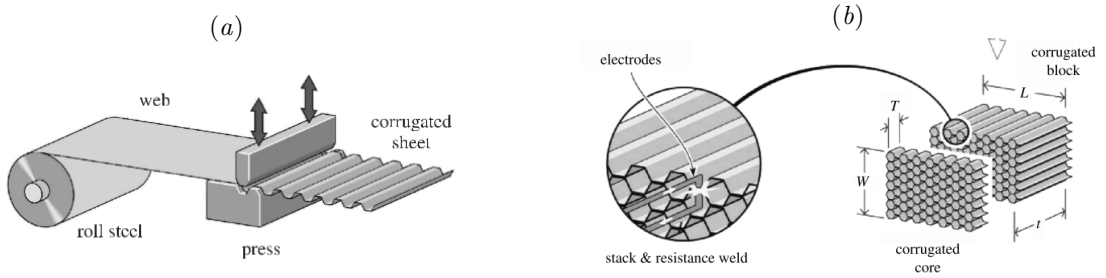


Figure 2. The corrugation process: (a) forming of corrugated sheet; (b) stacking and bonding of the corrugated sheets. Adapted from Wadley (2006).

3. Mechanics of energy absorption

The uniaxial compressive response of a cellular metal, sketched in figure 3, has three regimes that are common to all cellular solids (Gibson and Ashby, 1997; Altenbach and Öchsner, 2010). In the first the cell walls bend or stretch elastically and the response is reversible. When the stress reaches $\sigma_{\rm pl}$ the walls begin to collapse by plastic buckling and hinging, in bands that spread through the material (figure 4), and the stress then remains close to $\sigma_{\rm pl}$ over a large range of strain. Finally, opposing cell walls come into contact, the stress rises steeply towards that of the parent solid and little further energy can be absorbed.

The R–P–P–L idealisation (dashed line in figure 3) neglects the elastic strains, treating the material as rigid until $\sigma = \sigma_{\rm pl}$, perfectly plastic at constant stress up to $\varepsilon_{\rm D}$, and rigid again thereafter. Since the elastic strain at the end of the first regime is typically below 0.01, whereas $\varepsilon_{\rm D}$ is typically 0.6–0.8, the error is small. The energy absorbed per unit volume, and the total energy absorbed by a layer of area A and thickness h, are then (figure 5)

$$W_{\rm v} = \sigma_{\rm pl}\,\varepsilon_{\rm D}, \qquad E_{\rm abs} = F_{\rm max}\,\delta_{\rm D} = \sigma_{\rm pl}\varepsilon_{\rm D}Ah, \tag{3.1}$$

where $F_{\rm max} = \sigma_{\rm pl}A$ and $\delta_{\rm D} = \varepsilon_{\rm D}h$. A long plateau is thus the key to efficient absorption. Equally important for protection is that, until the foam locks, the force transmitted to the protected object cannot exceed $F_{\rm max}$.

4. Protection against low-velocity impact

4.1. Damage tolerance and the design plateau stress

Impact loads arise from accidental drops, from the deliberate landing of air-dropped loads, from vehicle collisions and from projectiles. The vulnerability of the protected object is expressed by its

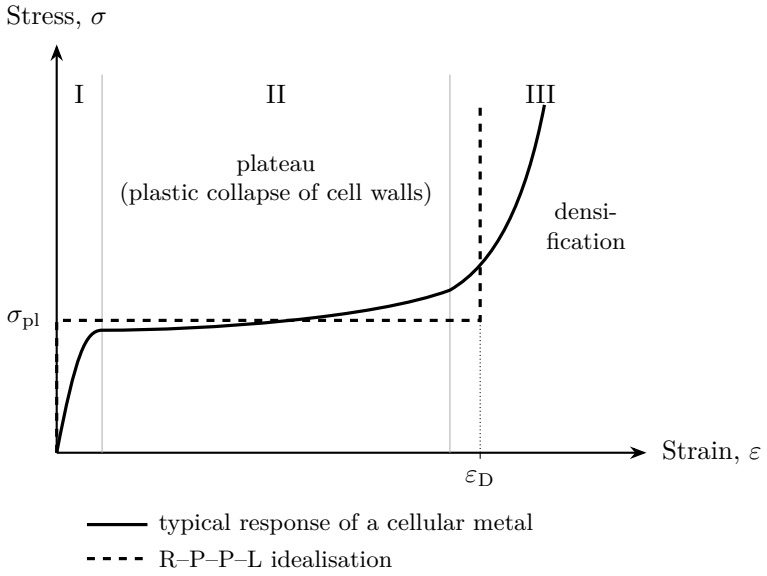


Figure 3. Compressive stress–strain response of a cellular metal (solid line) and its rigid, perfectly plastic, locking (R–P–P–L) idealisation (dashed line). The regimes are I, linear elastic; II, plateau; III, densification.

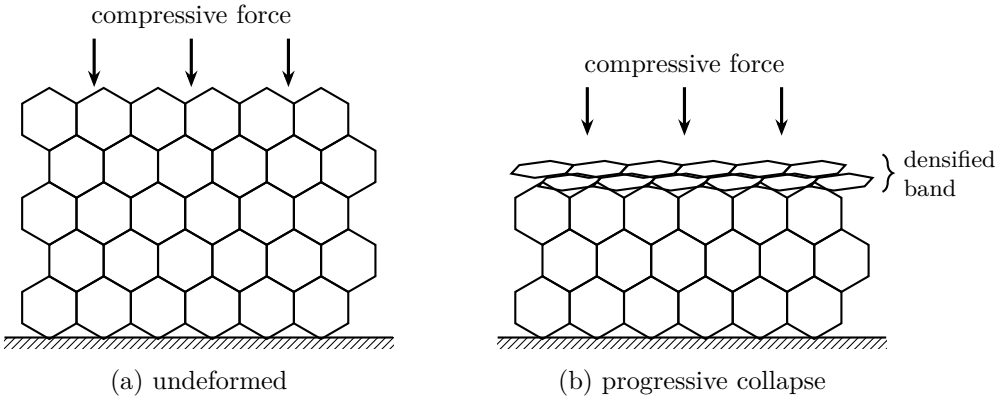


Figure 4. In-plane compression of a hexagonal honeycomb: (*a*) undeformed cells; (*b*) localised collapse forms a densified band that spreads through the core as compression continues.

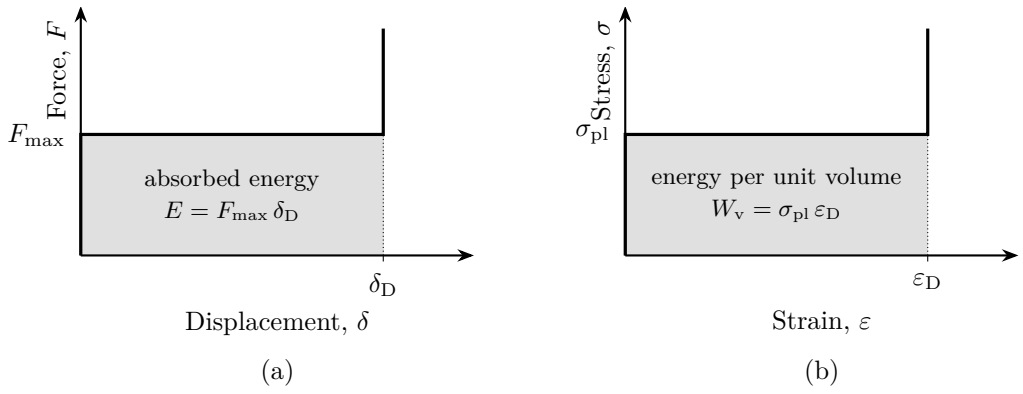


Figure 5. Energy absorbed by an R–P–P–L foam: (*a*) total energy, given by the area under the force–displacement curve; (*b*) energy per unit volume, given by the area under the stress–strain curve.

Table 2. Damage tolerance of common objects (Ashby et al., 2000).

Object	a^*
Human body	5–8
Delicate instruments	15–25
Optical equipment	25–40
Computer displays	40–60
Human head (36 ms contact)	55–60
Audio and television equipment	60–85
Household appliances	85–115
Machine tools, engines, chassis	115–150

Table 3. Typical impact velocities (Ashby et al., 2000).

Condition	v ($\mathrm{m\,s^{-1}}$)
Free fall from 1 m	4.4
High-drag parachute	7
Low-drag parachute	13
Car side impact	13.8
Car front impact	15.6

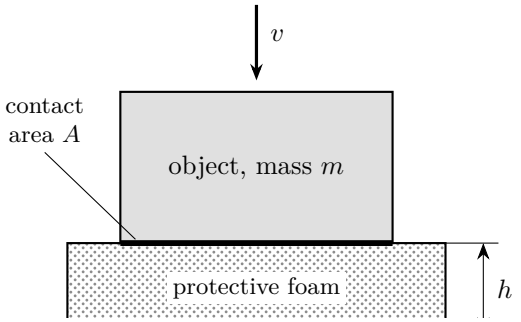


Figure 6. An object of mass m protected by a layer of foam of thickness h over a contact area A.

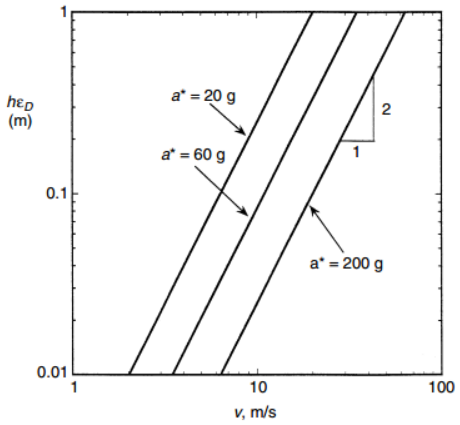


Figure 7. Required stroke $h\varepsilon_\mathrm{D}$ as a function of impact velocity at constant damage tolerance a^*. Reproduced from Ashby et al. (2000).

damage tolerance, or limiting g-factor, a^*: the largest deceleration, in multiples of the gravitational acceleration g, that it can sustain without damage (Ashby et al., 2000). Representative values, and typical impact velocities, are given in tables 2 and 3.

Consider an object of mass m resting on a layer of foam of thickness h over a contact area A (figure 6). For the deceleration to remain below a^*g, the transmitted force $\sigma_\mathrm{pl}A$ must not exceed ma^*g, so that the plateau stress of the foam must satisfy

$$\sigma_\mathrm{pl} \le \sigma^*_\mathrm{pl} = \frac{m\,a^*g}{A}. \tag{4.1}$$

The foam must also absorb the whole of the kinetic energy $E_\mathrm{k} = \frac{1}{2}mv^2$ before it locks. From (3.1) this requires

$$h = \frac{mv^2}{2\,\sigma_\mathrm{pl}\varepsilon_\mathrm{D}A}. \tag{4.2}$$

When $\sigma_\mathrm{pl} = \sigma^*_\mathrm{pl}$, (4.2) reduces to $h\varepsilon_\mathrm{D} = v^2/(2a^*g)$, which states simply that the usable stroke must equal the stopping distance at the limiting deceleration. Ashby et al. (2000) present this relation as a design chart (figure 7).

Table 4. Admissible metallic foams for the worked example, with W_m evaluated from (4.3) for $\varepsilon_\mathrm{D} = 0.8$. The Al–SiC (0.27) foam is selected.

Foam	ρ_0 (kg m^{-3})	σ_pl (MPa)	W_m (kJ kg^{-1})	h (mm)
Al–SiC (0.07)	70	0.044	0.50	1450
Al–SiC (0.16)	160	0.27	1.35	236
Al–SiC (0.27)	240	1.0	3.33	63.8

4.2. Material indices and selection

Many foams satisfy (4.1), and the best choice depends on the design objective (Ashby et al., 2000). A protective layer of least volume should maximise W_v; one of least mass should maximise the energy absorbed per unit mass; and one of least cost should maximise the energy absorbed per unit cost. These indices are

$$W_\mathrm{v} = \sigma_\mathrm{pl}\varepsilon_\mathrm{D}, \qquad W_\mathrm{m} = \frac{\sigma_\mathrm{pl}\varepsilon_\mathrm{D}}{\rho_0}, \qquad W_\mathrm{c} = \frac{\sigma_\mathrm{pl}\varepsilon_\mathrm{D}}{C_\mathrm{m}\,\rho_0}, \tag{4.3}$$

where ρ_0 is the density of the foam and C_m its cost per unit mass. Figure 8 shows the three indices plotted against σ_pl for the foams in the CES EduPack level-3 database (Granta Design Ltd, 2015), with the metallic (aluminium and aluminium–SiC) foams labelled. The resulting procedure, summarised in figure 9, has the merit that every quantity in it can be traced either to the specification of the object or to a published material property.

4.3. A worked example

As an illustration, consider an audio unit of mass $m = 5\,\mathrm{kg}$ that is to be delivered by high-drag parachute, resting on its feet, of total contact area $A = 2.4 \times 10^{-3}\,\mathrm{m}^2$, on a pad of metal foam whose mass is to be minimised. Tables 2 and 3 give $a^* = 60$ and $v = 7\,\mathrm{m\,s^{-1}}$, and (4.1) then gives $\sigma^*_\mathrm{pl} = 1.23\,\mathrm{MPa}$. A vertical line at this stress on figure 8(b) leaves three admissible metallic foams, compared in table 4. The Al–SiC (0.27) foam, taken at the lower end of its density range ($\rho_0 = 240\,\mathrm{kg\,m^{-3}}$, $\sigma_\mathrm{pl} \approx 1\,\mathrm{MPa}$), has the plateau stress closest to σ^*_pl and the largest W_m, and limits the deceleration to $\sigma_\mathrm{pl}A/mg \approx 49$. Equation (4.2) gives a thickness $h = 63.8\,\mathrm{mm}$ and a pad mass $\rho_0 Ah$ of only 37 g; a factor of safety of two on the stroke raises the thickness to 128 mm. The dependence of the required thickness on the impact velocity is shown in figure 10.

5. Impact of a rigid mass on a foam column

The procedure of § 4 assumes that the foam crushes uniformly at σ_pl. At higher impact velocities inertia becomes important: crushing localises at the impacted end and travels through the foam as a plastic compaction front, or shock (Reid and Peng, 1997; Harrigan et al., 2010). This section considers the canonical one-dimensional problem and compares two methods of solution that are in common use.

5.1. Shock relations for an R–P–P–L foam

On impact, an elastic precursor first travels through the foam at $c_\mathrm{el} = (E/\rho_0)^{1/2}$, where E is the Young modulus of the foam. It imparts a negligible particle velocity and is not represented in the R–P–P–L model (Ashby et al., 2000). It is followed by a plastic compaction front (figure 12), ahead of which the foam is at rest at density ρ_0 and stress σ_pl. Behind the front the foam is locked at strain ε_D and moves with the striker velocity v; its density is

$$\rho_\mathrm{D} = \frac{\rho_0}{1 - \varepsilon_\mathrm{D}}. \tag{5.1}$$

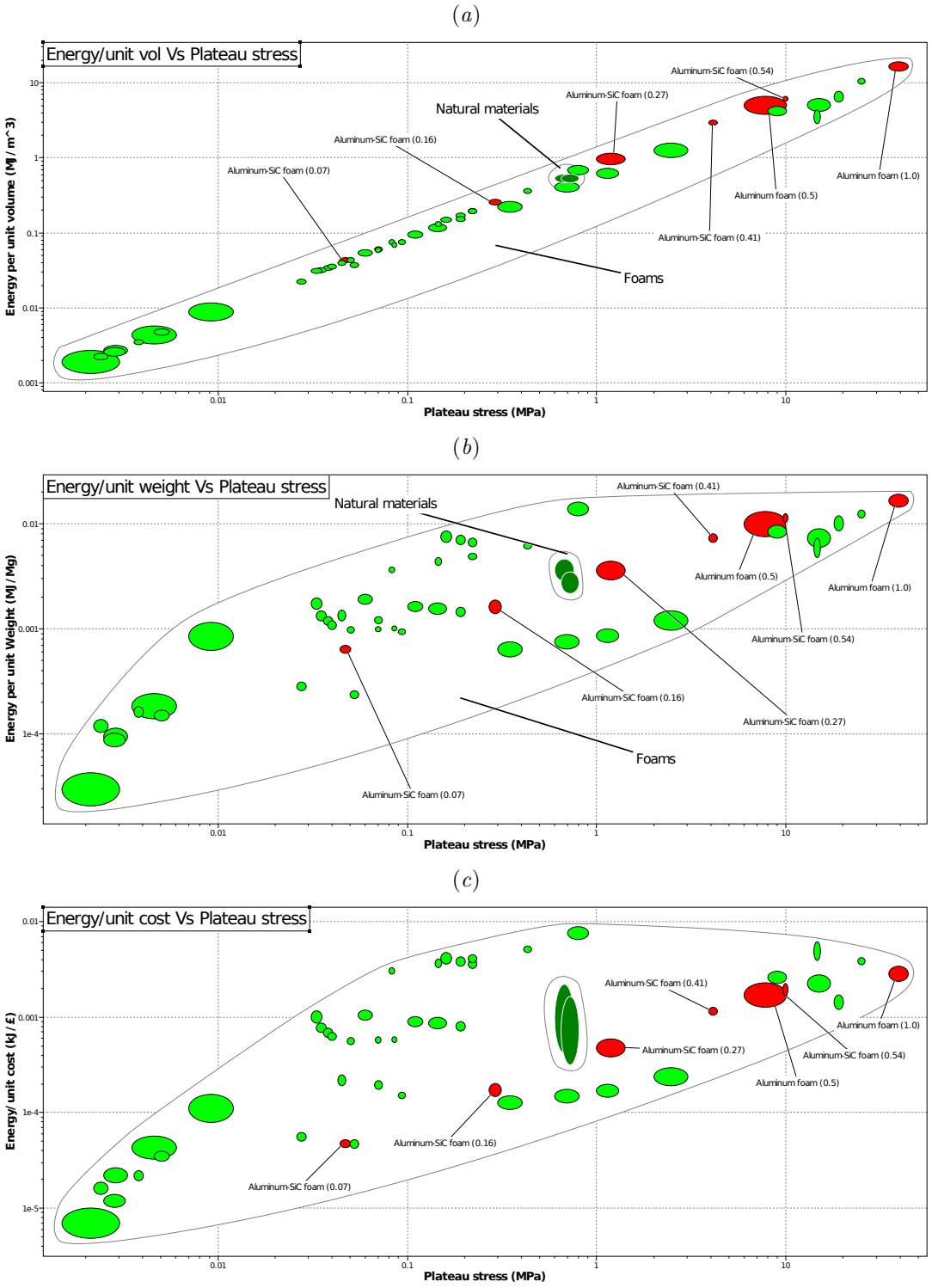


Figure 8. Energy absorbed per unit (a) volume, (b) mass and (c) cost, as functions of plateau stress, for foams in the CES EduPack 2015 level-3 database (Granta Design Ltd, 2015). Labelled points are metallic foams; the number in parentheses is the nominal density in $\mathrm{Mg\,m^{-3}}$.

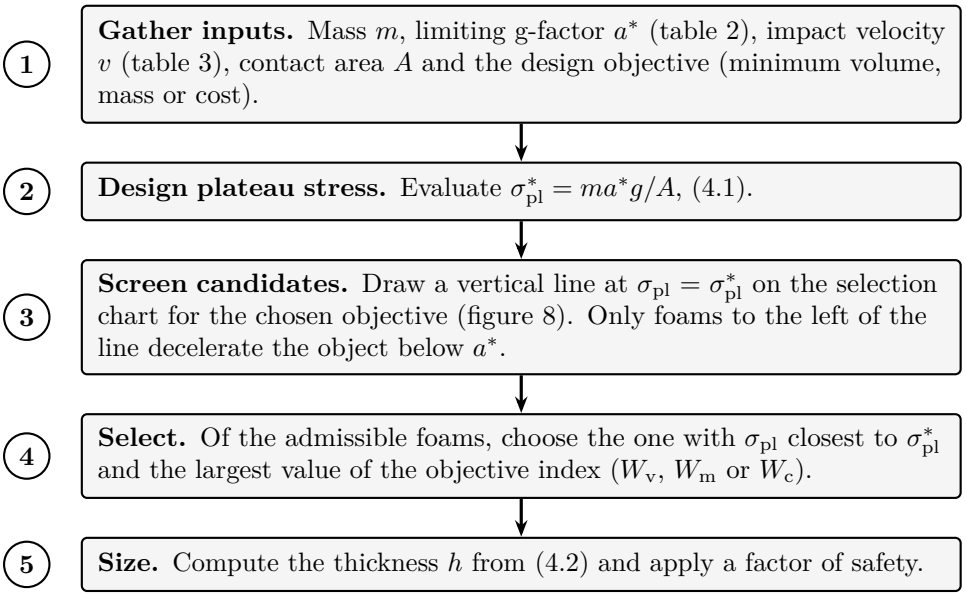


Figure 9. Procedure for the selection and sizing of cellular-metal impact protection.

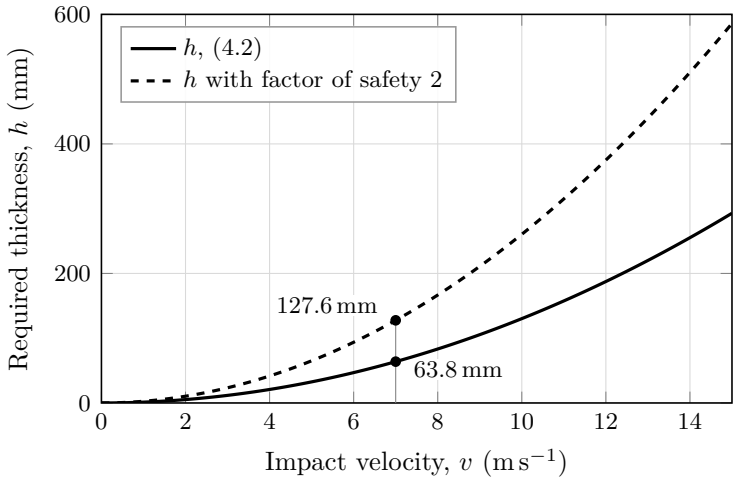


Figure 10. Thickness of Al–SiC (0.27) foam required for the worked example ($m = 5\,\mathrm{kg}$, $A = 2.4\times10^{-3}\,\mathrm{m}^2$, $\sigma_{\mathrm{pl}} = 1\,\mathrm{MPa}$, $\varepsilon_{\mathrm{D}} = 0.8$) as a function of impact velocity.

Conservation of mass and momentum across the front gives its speed relative to the undeformed foam, the stress behind it and the tangent modulus of the Rayleigh line (figure 11) (Reid and Peng, 1997; Harrigan et al., 2010):

$$c_{\mathrm{pl}} = \frac{v}{\varepsilon_{\mathrm{D}}} = \left(\frac{\sigma_{\mathrm{D}} - \sigma_{\mathrm{pl}}}{\rho_0 \varepsilon_{\mathrm{D}}}\right)^{1/2}, \qquad \sigma_{\mathrm{D}} = \sigma_{\mathrm{pl}} + \frac{\rho_0}{\varepsilon_{\mathrm{D}}}\, v^2, \qquad E_{\mathrm{t}} = \frac{\sigma_{\mathrm{D}} - \sigma_{\mathrm{pl}}}{\varepsilon_{\mathrm{D}}}. \tag{5.2}$$

The stress behind the front, which is the stress transmitted to the striker, therefore exceeds the quasi-static plateau stress by the dynamic contribution $\rho_0 v^2/\varepsilon_{\mathrm{D}}$. The length x of the compacted region grows at the rate

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{1-\varepsilon_{\mathrm{D}}}{\varepsilon_{\mathrm{D}}}\, v, \qquad \text{with} \qquad \frac{\rho_0}{\rho_{\mathrm{D}}} = 1 - \varepsilon_{\mathrm{D}} = \frac{x}{x+u}, \tag{5.3}$$

where u is the displacement of the striker (Harrigan et al., 2010).

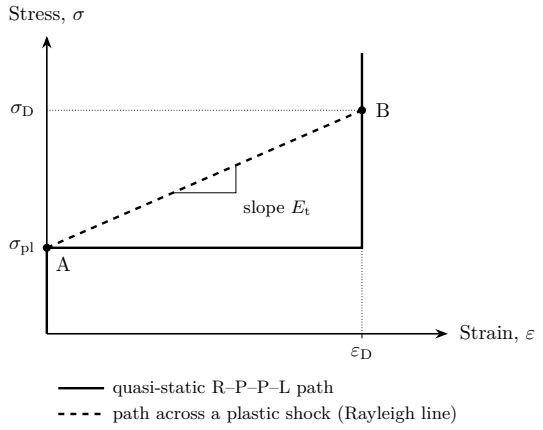


Figure 11. Stress–strain paths. Under quasi-static loading the material follows the R–P–P–L path; across a plastic shock it passes from state A to state B along the Rayleigh line, of slope E_{t}.

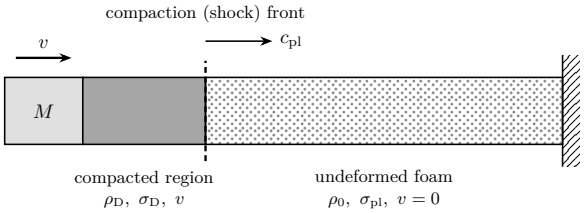


Figure 12. A compaction front travelling through a foam column struck by a mass M, separating compacted from undeformed material.

5.2. Formulation

A rigid mass M strikes, at velocity v_0, a foam column of initial length l_0, cross-sectional area A_0 and density ρ_0 whose far end is rigidly supported (figure 13). At time t the striker has moved a distance u at velocity v, and a compacted region of length x moves with it. Crushing stops when $v = 0$, and the column is just sufficient, with $l_0 = l_{\mathrm{crit}}$, when the striker comes to rest at the moment the whole column locks, that is when $u = u_{\max} = \varepsilon_{\mathrm{D}} l_{\mathrm{crit}}$.

5.3. Conservation of momentum

The system considered is the striker together with the compacted foam that moves with it. The only external force acting on it is $\sigma_{\mathrm{pl}} A_0$, exerted by the stationary foam ahead of the front, so that between t and $t + \delta t$ (Harrigan et al., 2010)

$$(M + \rho_{\mathrm{D}} A_0 (x + \delta x))(v + \delta v) - (M + \rho_{\mathrm{D}} A_0 x) v = -\sigma_{\mathrm{pl}} A_0 \, \delta t. \tag{5.4}$$

Eliminating x and δx with (5.3), and noting that $\rho_{\mathrm{D}} x = \rho_0 u / \varepsilon_{\mathrm{D}}$, gives the equation of motion

$$v \frac{\mathrm{d}v}{\mathrm{d}u} = - \frac{\sigma_{\mathrm{pl}} + \dfrac{\rho_0}{\varepsilon_{\mathrm{D}}} v^2}{\dfrac{M}{A_0} + \dfrac{\rho_0}{\varepsilon_{\mathrm{D}}} u}. \tag{5.5}$$

Equation (5.5) is linear in v^2. Writing $K(u) = M/A_0 + \rho_0 u / \varepsilon_{\mathrm{D}}$, the quantity $(v^2 + \sigma_{\mathrm{pl}} \varepsilon_{\mathrm{D}} / \rho_0) K^2$ is invariant (appendix A), whence

$$v^2 = \left(v_0^2 + \frac{\sigma_{\mathrm{pl}} \varepsilon_{\mathrm{D}}}{\rho_0} \right) \left(\frac{M/A_0}{M/A_0 + \rho_0 u / \varepsilon_{\mathrm{D}}} \right)^2 - \frac{\sigma_{\mathrm{pl}} \varepsilon_{\mathrm{D}}}{\rho_0}. \tag{5.6}$$

Setting $v = 0$ at $u = u_{\max}$ gives the impact velocity that just crushes a column of length l_0, and conversely the critical length that arrests a striker of velocity v_0:

$$v_0 = \left\{ \frac{\sigma_{\mathrm{pl}} \varepsilon_{\mathrm{D}}}{\rho_0} \left[\left(1 + \frac{\rho_0 A_0 u_{\max}}{M \varepsilon_{\mathrm{D}}} \right)^2 - 1 \right] \right\}^{1/2}, \tag{5.7}$$

$$l_{\mathrm{crit}}^{\mathrm{mom}} = \frac{M}{\rho_0 A_0} \left[(1 + \phi)^{1/2} - 1 \right], \qquad \phi \equiv \frac{\rho_0 v_0^2}{\sigma_{\mathrm{pl}} \varepsilon_{\mathrm{D}}}. \tag{5.8}$$

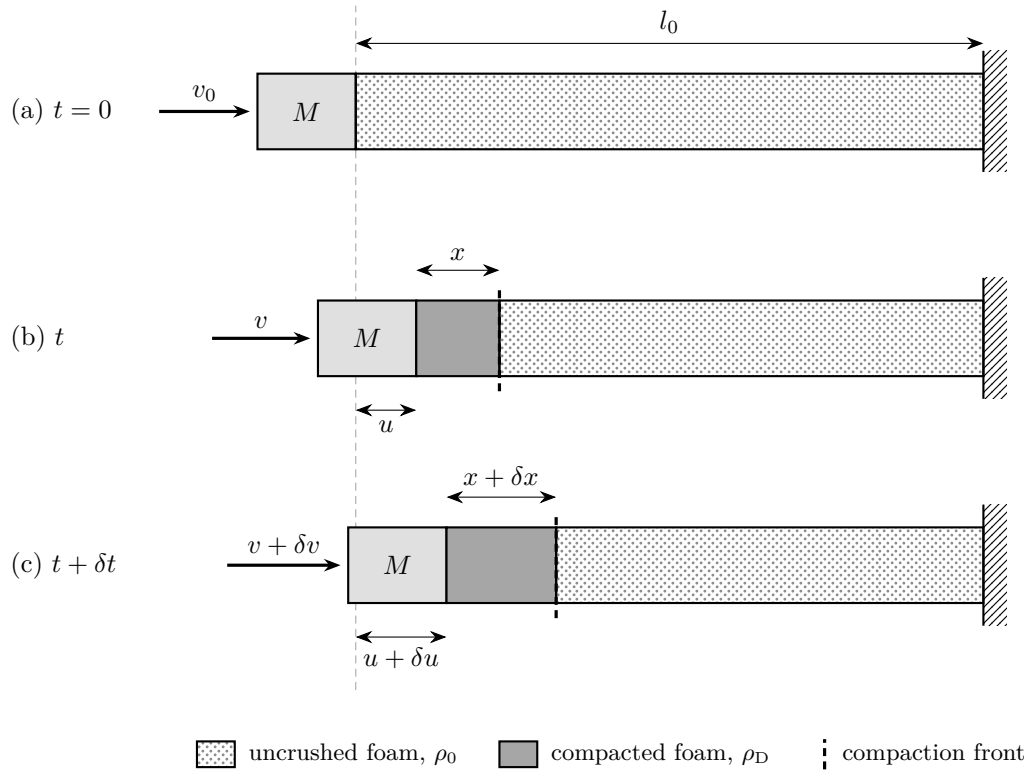


Figure 13. One-dimensional impact of a rigid mass M on an R–P–P–L foam column: (a) at first contact; (b) at time t; (c) at time $t + \delta t$.

The parameter ϕ is the ratio of the dynamic to the plastic contribution to σ_D at first contact. Substituting (5.6) into (5.2) gives the stress behind the front as the striker advances,

$$\sigma_D(u) = \left(\sigma_{pl} + \frac{\rho_0}{\varepsilon_D}\, v_0^2\right)\left(\frac{M/A_0}{M/A_0 + \rho_0 u/\varepsilon_D}\right)^2, \tag{5.9}$$

which takes its largest value, $\sigma_{pl}(1+\phi)$, at first contact and decays as compacted foam accumulates ahead of the striker.

5.4. Energy balance

A common alternative is to suppose that the kinetic energy lost by the striker and the compacted foam is dissipated only by plastic work at σ_{pl}, that is by the area under the R–P–P–L curve in figure 11. Between t and $t + \delta t$ this gives

$$(M + \rho_D A_0 x) v\, \delta v + \tfrac{1}{2}\rho_D A_0 v^2\, \delta x = -\sigma_{pl} A_0\, \delta u, \tag{5.10}$$

and hence, with (5.3),

$$v\frac{\mathrm{d}v}{\mathrm{d}u} = -\frac{\sigma_{pl} + \dfrac{\rho_0}{2\varepsilon_D}\, v^2}{\dfrac{M}{A_0} + \dfrac{\rho_0}{\varepsilon_D}\, u}. \tag{5.11}$$

Equations (5.11) and (5.5) differ only by the factor of one half on the inertial term. The invariant is now $v^2 K + 2\sigma_{pl} u$, so that

$$v^2 = \frac{v_0^2\, M/A_0 - 2\sigma_{pl} u}{M/A_0 + \rho_0 u/\varepsilon_D}, \tag{5.12}$$

Table 5. Critical length and initial stress behind the front for the parameters of figure 14.

v_0 (m s^{-1})	ϕ	$l_{\text{crit}}^{\text{mom}}$ (mm)	$l_{\text{crit}}^{\text{en}}$ (mm)	Ratio (5.15)	$\sigma_{\text{D}}(0)/\sigma_{\text{pl}}$
50	0.47	70.6	78.1	1.11	1.47
100	1.88	232	313	1.35	2.88
200	7.50	639	1250	1.96	8.50
300	16.9	1076	2813	2.61	17.9

and the corresponding full-crush velocity and critical length are

$$v_0 = \left(\frac{2A_0\sigma_{\text{pl}}u_{\max}}{M}\right)^{1/2}, \tag{5.13}$$

$$l_{\text{crit}}^{\text{en}} = \frac{Mv_0^2}{2A_0\sigma_{\text{pl}}\varepsilon_{\text{D}}} = \frac{M}{\rho_0 A_0}\,\frac{\phi}{2}. \tag{5.14}$$

Equation (5.13) expresses no more than the absorption of the initial kinetic energy at $W_{\text{v}} = \sigma_{\text{pl}}\varepsilon_{\text{D}}$, the assumption that also underlies (4.2).

5.5. Comparison of the two methods

From (5.8) and (5.14), the ratio of the critical lengths predicted by the two methods is

$$\frac{l_{\text{crit}}^{\text{en}}}{l_{\text{crit}}^{\text{mom}}} = \frac{\phi/2}{(1+\phi)^{1/2}-1} = \frac{1+(1+\phi)^{1/2}}{2} \geq 1. \tag{5.15}$$

The two predictions coincide as $\phi \to 0$, where $(1+\phi)^{1/2} - 1 \approx \phi/2$, and diverge as ϕ increases. The reason lies in the energy dissipated in the shock. Across the front the material follows the Rayleigh line A–B of figure 11, rather than the R–P–P–L path, and the triangle between the two paths, of area $\frac{1}{2}(\sigma_{\text{D}} - \sigma_{\text{pl}})\varepsilon_{\text{D}}$ per unit volume, represents additional dissipation that the energy balance omits (Reid and Peng, 1997; Harrigan et al., 2010). The energy method therefore over-predicts the length of foam required: it is conservative, but increasingly wasteful as the impact velocity rises. A convenient threshold is the velocity at which $\phi = 1$,

$$v^* = \left(\frac{\sigma_{\text{pl}}\varepsilon_{\text{D}}}{\rho_0}\right)^{1/2}. \tag{5.16}$$

Below v^* the two predictions differ by less than 21 % and a quasi-static design is adequate; above it, the effects of the shock should be included.

These results are illustrated in figures 14 and 15 and in table 5 for a representative closed-cell aluminium foam ($\rho_0 = 300\,\text{kg m}^{-3}$, $\sigma_{\text{pl}} = 2\,\text{MPa}$, $\varepsilon_{\text{D}} = 0.8$, so that $v^* \approx 73\,\text{m s}^{-1}$) struck by a mass of 1 kg over an area of $0.01\,\text{m}^2$. At $50\,\text{m s}^{-1}$ the two methods differ by 11 %; at $300\,\text{m s}^{-1}$ the energy balance requires 2.6 times as much foam as the momentum solution. At this speed the initial stress behind the front is almost eighteen times the plateau stress, which matters in its own right, since it is σ_{D} rather than σ_{pl} that is transmitted to the striker.

6. Protection against blast

6.1. Air-blast loading

The detonation of a high explosive converts almost all of the chemical energy released into a blast wave (Smith and Hetherington, 1994): a thin shell of highly compressed air expands from

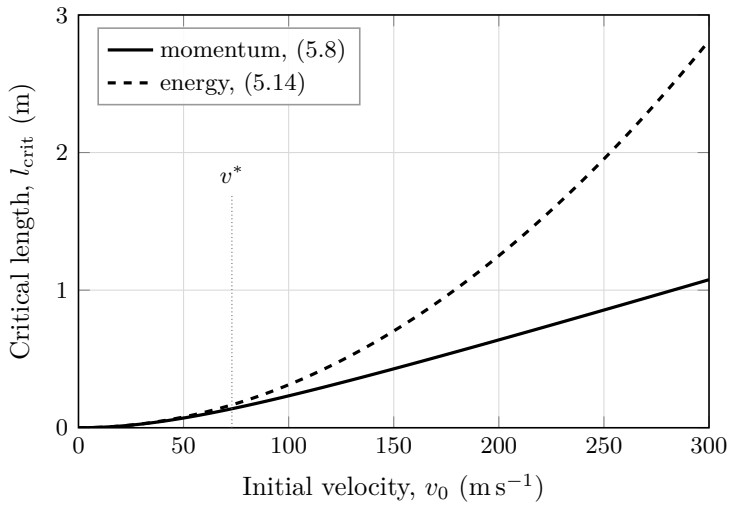


Figure 14. Critical length of foam required to arrest a striker of mass 1 kg ($A_0 = 0.01\,\mathrm{m}^2$, $\rho_0 = 300\,\mathrm{kg\,m^{-3}}$, $\sigma_{\mathrm{pl}} = 2\,\mathrm{MPa}$, $\varepsilon_{\mathrm{D}} = 0.8$), as predicted by conservation of momentum and by the energy balance.

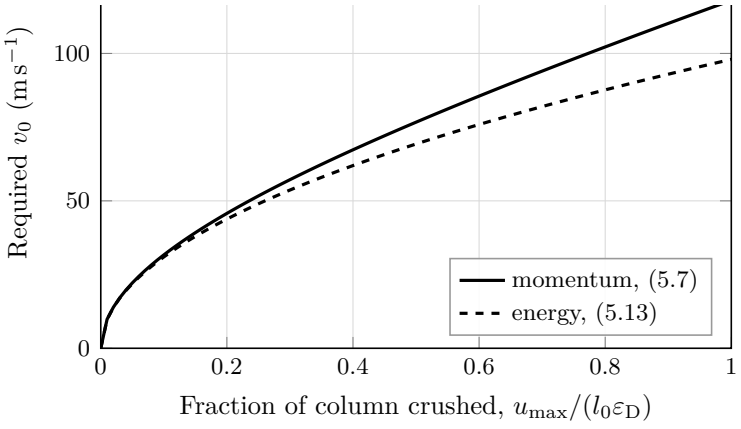


Figure 15. Impact velocity required to crush a given fraction of a column of length $l_0 = 0.3\,\mathrm{m}$; other parameters as in figure 14.

the charge and imposes a short pressure pulse $p(t)$ on any structure that it meets. When the pulse is short compared with the response time of the structure, it is characterised by its specific impulse,

$$I = \int_0^{t_{\mathrm{d}}} p(t)\,\mathrm{d}t, \tag{6.1}$$

where t_{d} is the duration of the positive phase. Hopkinson–Cranz (cube-root) scaling collapses data for different charge masses W, expressed as an equivalent mass of TNT, and stand-off distances R onto the scaled distance $Z = R/W^{1/3}$ (Smith and Hetherington, 1994; Karlos and Solomos, 2013). For a free air burst, Kinney and Graham (1985) give the scaled side-on impulse as

$$\frac{I_{\mathrm{s}}}{W^{1/3}} = \frac{0.067\,[1 + (Z/0.23)^4]^{1/2}}{Z^2\,[1 + (Z/1.55)^3]^{1/3}} \quad [\mathrm{bar\,ms\,kg^{-1/3}}], \tag{6.2}$$

with Z in $\mathrm{m\,kg^{-1/3}}$ and $1\,\mathrm{bar\,ms} = 100\,\mathrm{Pa\,s}$. Figure 16 shows the rapid decay of impulse with scaled distance, which is consistent with other compilations (Larcher, 2007).

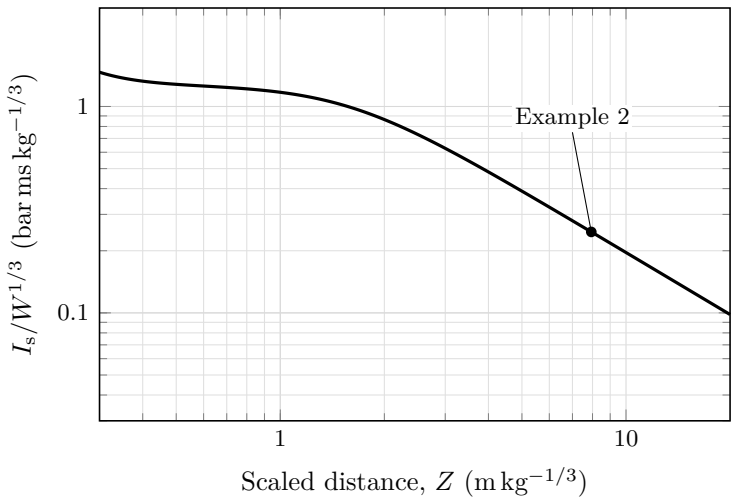


Figure 16. Scaled side-on impulse of a free air burst as a function of scaled distance, from (6.2) (Kinney and Graham, 1985).

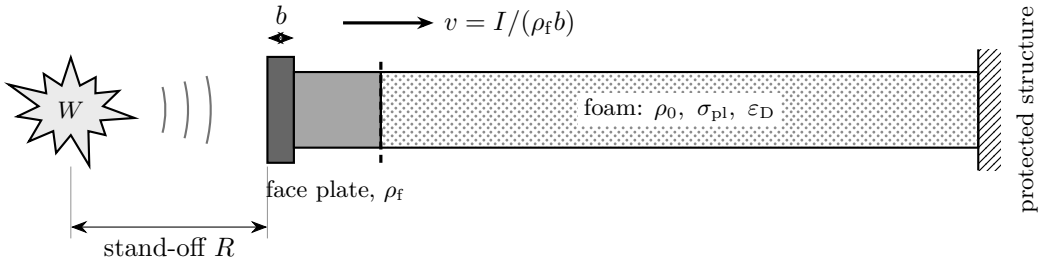


Figure 17. Sacrificial cladding: a face plate backed by a layer of foam, loaded by an air blast from a charge of mass W at stand-off R.

6.2. Sacrificial foam cladding

The protective arrangement of figure 17 consists of a face plate, of thickness b and density ρ_f, bonded to a layer of foam that is supported by the structure to be protected. Because the blast pulse is usually much shorter than the time taken to crush the foam, the impulse is transferred to the plate before the foam has deformed appreciably, and the plate acquires a velocity

$$v = \frac{I}{\rho_f b}, \tag{6.3}$$

and a kinetic energy per unit area $I^2/(2\rho_f b)$ (Ashby et al., 2000; Hanssen et al., 2002). The problem then reduces to that of § 5, with the plate as the striker and $M/A_0 = \rho_f b$. The energy method, (5.14), gives the length of foam crushed as

$$l_{crit} = \frac{I^2}{2\rho_f b\, \sigma_{pl}\varepsilon_D}. \tag{6.4}$$

Provided that the foam is longer than l_{crit}, the pressure transmitted to the structure is limited to σ_{pl}. Equation (6.4) also shows that, for a given impulse, a heavier plate reduces the length of foam required, since it receives less kinetic energy.

6.3. An example

Consider a steel face plate ($b = 5\,\text{mm}$, $\rho_f = 7850\,\text{kg m}^{-3}$, so that $\rho_f b = 39.3\,\text{kg m}^{-2}$) backed by the aluminium foam of § 5.5. Table 6 gives the plate velocity and the crushed length for

Table 6. Response of steel-faced aluminium-foam cladding ($b = 5\,\mathrm{mm}$, $\sigma_{\mathrm{pl}} = 2\,\mathrm{MPa}$, $\varepsilon_{\mathrm{D}} = 0.8$, $\rho_0 = 300\,\mathrm{kg\,m^{-3}}$) to free air bursts, using the side-on impulse of (6.2).

W (kg)	R (m)	Z ($\mathrm{m\,kg^{-1/3}}$)	I (Pa s)	v ($\mathrm{m\,s^{-1}}$)	E_{k}/A ($\mathrm{J\,m^{-2}}$)	l_{crit} (mm)
2	10	7.94	31	0.79	12	0.01
2	5	3.97	61	1.56	48	0.03
10	5	2.32	167	4.26	356	0.22
10	2	0.93	256	6.53	836	0.52
50	2	0.54	467	11.9	2784	1.74

several threats. For a charge of 2 kg at 10 m, $Z = 7.94\,\mathrm{m\,kg^{-1/3}}$ and $I = 31\,\mathrm{Pa\,s}$; only $12\,\mathrm{J\,m^{-2}}$ is transferred, and less than 0.1 mm of foam is crushed. Even for a charge of 50 kg at 2 m the plate velocity, $12\,\mathrm{m\,s^{-1}}$, is far below $v^* = 73\,\mathrm{m\,s^{-1}}$, so that $\phi < 0.03$, the shock enhancement is negligible and the energy method is accurate for side-on loading of this kind.

The crushed lengths are small because (6.2) gives the incident impulse of a free air burst. For cladding mounted on a surface the reflected impulse is the relevant quantity; it may be several times larger, and l_{crit} scales with I^2. Near-field and contact charges deliver much larger impulses still, for which the shock and the deformation of the face plate must be considered (Smith and Hetherington, 1994; Hanssen et al., 2002). The values in table 6 are therefore best read as a lower bound that illustrates the scaling, rather than as design values.

7. Comparison with thin-walled tubes

Thin-walled metal tubes are the other classical energy absorber. Under axial compression beyond a critical load the wall buckles into a sequence of progressive folds (figures 18 and 19), each of which dissipates energy at plastic hinge lines, much as the cell walls of a foam do. For a tube of radius r, wall thickness t_{w} and yield strength σ_{ys} the mean crushing load is approximately (Ashby et al., 2000)

$$F_{\mathrm{m}} = 4\pi\, r^{1/3} t_{\mathrm{w}}^{5/3}\, \sigma_{\mathrm{ys}}. \tag{7.1}$$

The load fluctuates about F_{m} as each fold forms and closes, until the folds lock at a compaction strain $\varepsilon_{\mathrm{D}}^{\mathrm{tube}}$ and stroke $\delta = l\,\varepsilon_{\mathrm{D}}^{\mathrm{tube}}$. With the effective relative density of the tube $\rho/\rho_{\mathrm{s}} \approx 2t_{\mathrm{w}}/r$, the energy absorbed per unit enclosed volume is

$$W_{\mathrm{v}}^{\mathrm{tube}} = \frac{F_{\mathrm{m}}\delta}{\pi r^2 l} = 4\left(\frac{t_{\mathrm{w}}}{r}\right)^{5/3} \sigma_{\mathrm{ys}}\, \varepsilon_{\mathrm{D}}^{\mathrm{tube}} = 2^{1/3}\left(\frac{\rho}{\rho_{\mathrm{s}}}\right)^{5/3} \sigma_{\mathrm{ys}}\, \varepsilon_{\mathrm{D}}^{\mathrm{tube}}. \tag{7.2}$$

For a foam made from the same solid, $\sigma_{\mathrm{pl}} \approx 0.3(\rho/\rho_{\mathrm{s}})^{3/2}\sigma_{\mathrm{ys}}$ (Gibson and Ashby, 1997; Ashby et al., 2000), so that $W_{\mathrm{v}}^{\mathrm{foam}} \approx 0.3(\rho/\rho_{\mathrm{s}})^{3/2}\sigma_{\mathrm{ys}}\,\varepsilon_{\mathrm{D}}^{\mathrm{foam}}$. If the densification strains are similar,

$$\frac{W_{\mathrm{v}}^{\mathrm{tube}}}{W_{\mathrm{v}}^{\mathrm{foam}}} \approx 4.2\left(\frac{\rho}{\rho_{\mathrm{s}}}\right)^{1/6}, \tag{7.3}$$

a ratio that depends only weakly on density, taking the values 2.9 at $\rho/\rho_{\mathrm{s}} = 0.1$ and 3.2 at 0.2. At equal relative density, therefore, an axially loaded tube absorbs about three times as much energy per unit volume as a foam, and sustains a much higher crushing load (figure 20). The foam nevertheless has three advantages. It is nearly isotropic, and so continues to absorb energy under oblique impact that would cause a tube to collapse globally by Euler buckling; it delivers an almost constant force, without the initial peak and periodic oscillation of the folding tube; and it can be shaped to fill an arbitrary volume.

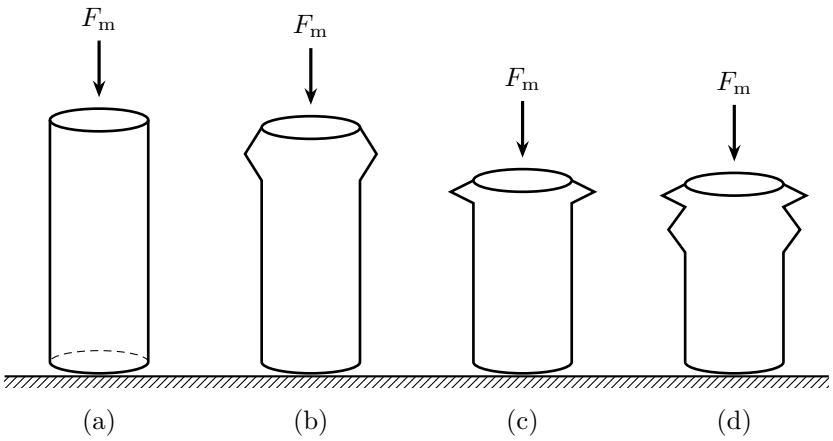


Figure 18. Progressive folding of a thin-walled cylinder under axial load F_m: (a) undeformed; (b) initiation of the first fold; (c) completion of the first fold; (d) initiation of the second fold.

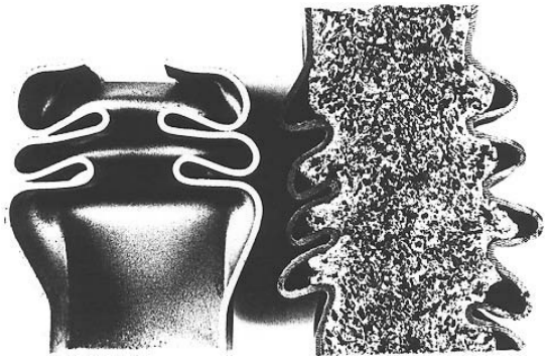


Figure 19. Axially crushed tubes, showing the progressive folding pattern and, in the foam-filled case, foam compressed into the cavities of the folds. Reproduced from Ashby et al. (2000).

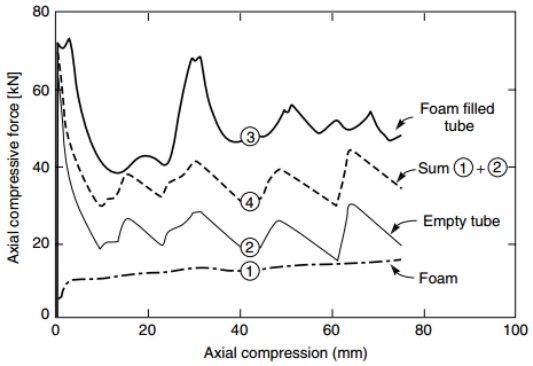


Figure 20. Load–deflection response of a foam, an empty tube and a foam-filled tube. Reproduced from Ashby et al. (2000).

Filling a tube with foam combines these advantages. The foam stabilises the wall, shortens the wavelength of the folds and is itself compressed into the fold cavities (figure 19), producing more plastic folds per unit length (Abramowicz and Wierzbicki, 1988; Hanssen et al., 1999). The energy absorbed by the filled tube then exceeds the sum of its parts by an interaction term,

$$W_{\mathrm{v}}^{\mathrm{filled}} = W_{\mathrm{v}}^{\mathrm{tube}} + W_{\mathrm{v}}^{\mathrm{foam}} + W_{\mathrm{v}}^{\mathrm{int}}, \tag{7.4}$$

which can be comparable to $W_{\mathrm{v}}^{\mathrm{foam}}$ (Hanssen et al., 1999). Foam filling thus raises the specific energy absorption of a tube, although it slightly reduces the usable stroke.

8. Finite-element illustration

To visualise the deformation modes described above, the compression was simulated of a thin-walled cylinder and of a block of aluminium honeycomb with a finite-element package integrated into a computer-aided design environment, using aluminium alloy 3003, which is typical of tubular absorbers and honeycomb cores. The analyses are qualitative and are not used to extract values of absorbed energy. The cylinder (figure 21a,b) was loaded axially with its ends fixed, representing a tube welded to end structures. A circumferential fold forms at mid-height, with incipient folds visible on either side of it, consistent with the progressive mode of figure 18. The honeycomb (figure 21c,d), compressed normal to the cell axis between fixed faces, develops

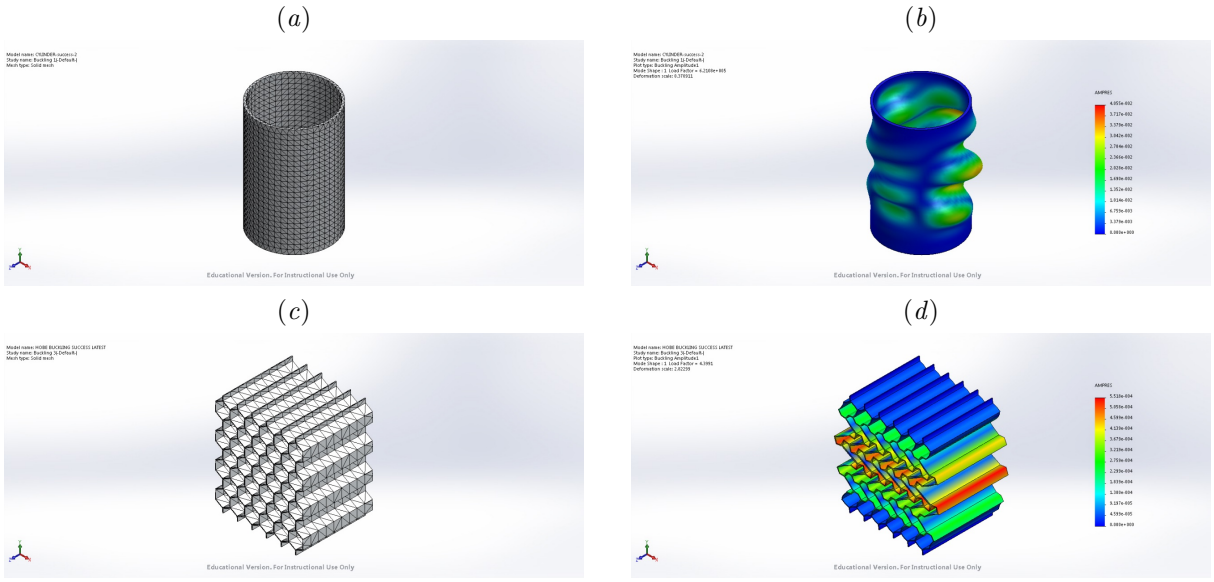


Figure 21. Finite-element illustration of the deformation modes of aluminium alloy 3003: (*a*) meshed thin-walled cylinder and (*b*) the same cylinder after axial compression, showing the formation of a fold; (*c*) meshed honeycomb (HOBE) block and (*d*) the same block after in-plane compression, showing cell collapse and densification.

localised cell collapse and a band of densified material, as sketched in figure 4. The detailed collapse pattern of the honeycomb was sensitive to the boundary conditions, as is well known for in-plane crushing (Gibson and Ashby, 1997).

9. Discussion

The treatment rests throughout on the R–P–P–L idealisation, which makes every result explicit and easily checked but neglects several effects. Real foams show a rising plateau and a gradual approach to densification (figure 3), so that ε_{D} must be defined consistently, for example by an energy-efficiency criterion; strain-rate sensitivity of the cell-wall material and the pressure of gas in closed cells raise σ_{pl} under dynamic loading; elastic precursors and partial unloading are ignored; and the properties of metallic foams are subject to considerable scatter, as the ranges in figure 8 indicate. Within these limitations the comparison of § 5.5 leads to a simple rule. Sizing by energy, as in (4.2) and (6.4), is adequate and conservative when $v_0 < v^*$. Above v^* the momentum solution (5.8) should be used, both to avoid over-design and, more importantly, because the transmitted stress σ_{D} then exceeds σ_{pl} by the factor $1 + \phi$; this may violate the damage-tolerance criterion (4.1) even when the foam is long enough. For blast, the correlation (6.2) applies to free air bursts; reflected and near-field loading require dedicated models of the blast load (Karlos and Solomos, 2013; Larcher, 2007) and, where the deformation of the face plate matters, numerical simulation.

10. Conclusions

A unified analytical treatment of the design of cellular-metal energy absorbers has been presented. For low-velocity impact, damage-tolerance data combined with charts of energy absorbed per unit volume, mass or cost lead to a simple procedure for selection and sizing; in the example considered, a pad of Al–SiC foam 64 mm thick and weighing 37 g suffices to protect a 5 kg unit landing at $7\,\mathrm{m\,s^{-1}}$.

For a rigid mass striking a foam column, conservation of momentum across the compaction front

and a global energy balance give equations of motion that differ only by a factor of one half on the inertial term. The critical lengths that they predict are in the ratio $[1 + (1 + \phi)^{1/2}]/2$, so that the energy balance is always conservative, and the difference becomes important above the characteristic velocity $v^* = (\sigma_{\mathrm{pl}}\varepsilon_{\mathrm{D}}/\rho_0)^{1/2}$, about $73\,\mathrm{m\,s^{-1}}$ for an aluminium foam of density $300\,\mathrm{kg\,m^{-3}}$. For sacrificial blast cladding, the face-plate velocities produced by realistic side-on impulses lie well below v^*, so that the energy method is appropriate and the crushed length scales as $I^2/\rho_{\mathrm{f}}b$.

At equal relative density, an axially loaded thin-walled tube absorbs about three times as much energy per unit volume as a foam. Foams offer isotropy, a nearly constant force and freedom of shape, and foam-filled tubes exploit a beneficial interaction between the two. The closed-form results are intended as a transparent first estimate in design and as benchmarks for numerical models.

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Data availability statement. All results can be reproduced from the closed-form expressions given in the paper. The scripts used to generate the figures are available as supplementary material.

A. Integration of the equations of motion

Let $w = v^2$ and $K(u) = M/A_0 + \rho_0 u/\varepsilon_{\mathrm{D}}$, so that $\mathrm{d}K/\mathrm{d}u = \rho_0/\varepsilon_{\mathrm{D}}$ and $v\,\mathrm{d}v/\mathrm{d}u = \frac{1}{2}\,\mathrm{d}w/\mathrm{d}u$.

For the momentum formulation, (5.5) becomes $K\,\mathrm{d}w/\mathrm{d}u = -2\sigma_{\mathrm{pl}} - 2(\rho_0/\varepsilon_{\mathrm{D}})w$. Writing $y = w + \sigma_{\mathrm{pl}}\varepsilon_{\mathrm{D}}/\rho_0$ gives $\mathrm{d}y/y = -2\,\mathrm{d}K/K$, so that yK^2 is constant. Evaluating the constant at $u = 0$, where $w = v_0^2$ and $K = K_0 = M/A_0$, yields (5.6). Setting $w = 0$ gives $(K/K_0)^2 = 1 + \phi$, from which (5.7) and (5.8) follow with $u_{\max} = \varepsilon_{\mathrm{D}} l_{\mathrm{crit}}$.

For the energy formulation, (5.11) becomes $K\,\mathrm{d}w/\mathrm{d}u + (\rho_0/\varepsilon_{\mathrm{D}})w = -2\sigma_{\mathrm{pl}}$, that is $\mathrm{d}(wK)/\mathrm{d}u = -2\sigma_{\mathrm{pl}}$. Hence $wK = v_0^2 M/A_0 - 2\sigma_{\mathrm{pl}}u$, which is (5.12). At $w = 0$, $u_{\max} = Mv_0^2/(2A_0\sigma_{\mathrm{pl}})$, independent of ρ_0, which gives (5.13).

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